Majority Element

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The problem

- An element occurs in more than half the array; find it (Boyer-Moore voting).

Intuition

- Boyer-Moore voting: pair off different values; the majority survives the cancellation.

Approach

- 1. Keep a candidate and a count
- 2. Same value increments, different decrements
- 3. When the count hits zero, adopt the current value
- 4. The final candidate is the answer

Java solution

```java
public class MajorityElement {
    public static int majorityElement(int[] a) {
        int cand = 0, count = 0;
        for (int x : a) {
            if (count == 0) cand = x;
            count += (x == cand) ? 1 : -1;
        }
        return cand;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Boyer-Moore voting: pair off different values; the majority survives the cancellation.
- Complexity: O(n)